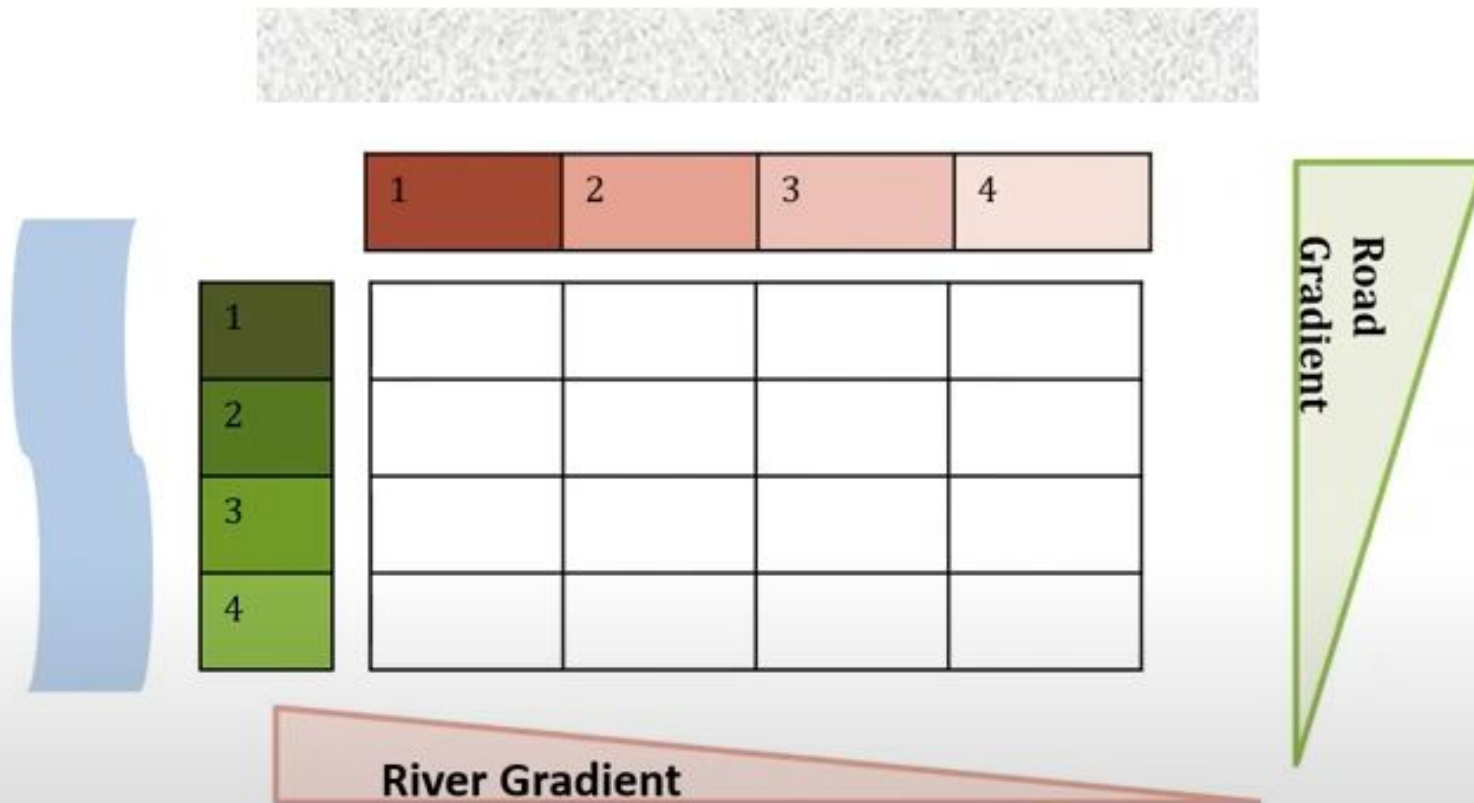


Latin Square Design

Latin Square Design

- RCBD – minimizes the contribution of one source of variation other than treatment
- Latin square design can handle **two** sources of variation that occur in a gradient
- Every treatment occurs only once in each row and column
- For example
- A field has **river** on one side and a **road** on the other side

Latin Square Design



Latin Square Design

- Can have a maximum of only **16** plots
- A, B, C, D are types of treatments
- Each treatment appears only once in row (**river gradient, A**) and column (**road gradient, B**)

		River gradient			
		1	2	3	4
Road gradient	1	D	A	B	C
	2	A	D	C	B
	3	C	B	A	D
	4	B	C	D	A

Latin Square Design

- Example
- Effects of fertilizer dose on the fresh weight of plants
- Experimental design – Latin Square Design
- Independent variable – fertilizer doses
- Treatments – different doses of fertilizer – A, B, C and D
- Dependent (response) variable – fresh weight of plants

Latin Square Design

- Fresh weight (g) of plants under different doses of fertilizers

	1	2	3	4
1	34.45 D	64.16 A	53.13 B	49.02 C
2	66.42 A	31.31 D	49.60 C	51.94 B
3	52.54 C	54.94 B	62.73 A	25.84 D
4	58.05 B	53.73 C	29.33 D	56.79 A

Latin Square Design

- Table for ANOVA calculation
- According to gradients

	1	2	3	4	Row total	Row $\sum x^2$
1	34.45 D	64.16 A	53.13 B	49.02 C	200.76	10529.07
2	66.42 A	31.31 D	49.60 C	51.94 B	199.27	10549.86
3	52.54 C	54.94 B	62.73 A	25.84 D	196.05	10381.61
4	58.05 B	53.73 C	29.33 D	56.79 A	197.90	10342.07
Column total	211.46	204.14	194.79	183.59	$\sum x_{To}$ 793.98	
Column sum $\sum x^2$	11728.67	11002.14	10078.26	8993.53	$\sum x_{To}^2$ 41802.61	

Latin Square Design

- Treatment observations are listed separately
- Treatment A = $\sum x = 250.1$ $\sum x^2 = 15688.28$
- Treatment B = $\sum x = 218.06$ $\sum x^2 = 11908.77$
- Treatment C = $\sum x = 204.89$ $\sum x^2 = 10510.48$
- Treatment D = $\sum x = 120.93$ $\sum x^2 = 3695.073$

Latin Square Design

- $SS_{\text{Total}} = SS_{\text{Treatment}} + SS_{\text{GradientA}} + SS_{\text{GradientB}} + SS_{\text{ExperimentalError}}$
- For the sake of simplicity
- $SS_{T_o} = SS_t + SS_A + SS_B + SS_e$
- Gradient A – road gradient, rows
- Gradient B – river gradient, columns
- **Calculate CT**
- $CT = (\sum x_{T_o})^2 / (n_{T_o}) = (793.98)^2 / 16$
 $= 630404.24 / 16$
- ~~**CT = 39400.265**~~

Latin Square Design

- Calculate SS_{T_o}
- $SS_{T_o} = \sum x_{T_o}^2 - CT = 41802.61 - 39400.265$
- **$SS_{T_o} = 2402.345$**
- Calculate SS_t
- $SS_t = \{[(\sum x_{t1})^2/n_{t1}] + [(\sum x_{t2})^2/n_{t2}] + [(\sum x_{t3})^2/n_{t3}] + [(\sum x_{t4})^2/n_{t4}]\} - CT$
- $SS_t = \{[(250.1)^2/4] + [(218.06)^2/4] + [(204.89)^2/4] + [(120.93)^2/4]\} - 39400.265$
- **$SS_t = 2275.77$**

Latin Square Design

- Calculate SS_A (road gradient, rows)
 - $SS_A = \{[(\sum x_{A1})^2/n_{A1}] + [(\sum x_{A2})^2/n_{A2}] + [(\sum x_{A3})^2/n_{A3}] + [(\sum x_{A4})^2/n_{A4}]\} - CT$
 - $SS_A = \{[(200.76)^2/4] + [(199.27)^2/4] + [(196.05)^2/4] + [(197.9)^2/4]\} - 39400.265$
 - **$SS_A = 3.02$**
-
- Calculate SS_B (river gradient, columns)
 - $SS_B = \{[(\sum x_{B1})^2/n_{B1}] + [(\sum x_{B2})^2/n_{B2}] + [(\sum x_{B3})^2/n_{B3}] + [(\sum x_{B4})^2/n_{B4}]\} - CT$
 - $SS_B = \{[(211.46)^2/4] + [(204.14)^2/4] + [(194.79)^2/4] + [(183.59)^2/4]\} - 39400.265$
 - **$SS_B = 108.96$**

Latin Square Design

- Calculate SS_e
- $SS_{To} = SS_t + SS_A + SS_B + SS_e$
- $SS_e = SS_{To} - (SS_t + SS_A + SS_B)$
- $SS_e = 2402.345 - (2275.77 + 3.02 + 108.96)$
- **$SS_e = 14.595$**

Latin Square Design

- **Calculate df**
- Total df = $n_{T_0} - 1 = 16 - 1 = 15$
- Treatment df = $n - 1 = 4 - 1 = 3$
- Gradient A df = $n - 1 = 4 - 1 = 3$
- Gradient B df = $n - 1 = 4 - 1 = 3$
- Error df = $(n - 1)(n - 2) = (4 - 1)(4 - 2) = 6$
- n = number of columns = number of rows

Latin Square Design

- Calculate MS
- $MS_t = SS_t/df_t = 2275.77/3$
- **$MS_t = 758.59$**
- $MS_A = SS_A/df_A = 3.02/3$
- **$MS_A = 1.007$**
- $MS_B = SS_B/df_B = 108.96/3$
- **$MS_B = 36.32$**
- $MS_e = SS_e/df_e = 14.595/6$
- **$MS_e = 2.433$**

Latin Square Design

- Calculate F
- $F_t = MS_t/MS_e = 758.59/2.433$
- **$F_t = 311.79$**
- $F_A = MS_A/MS_e = 1.003/2.433$
- **$F_A = 0.412$**
- $F_B = MS_B/MS_e = 36.32/2.433$
- **$F_B = 14.928$**

Latin Square Design

- **ANOVA Table**

Source of variation	df	SS	MS	F	P
Treatment	3	2275.77	758.59	311.79	< 0.001
Gradient A	3	3.02	1.003	0.412	> 0.05
Gradient B	3	108.96	36.32	14.928	< 0.01
Error	6	14.595	2.433		
Total	15	2402.35			

- Tabled F values = 4.76 at p 0.05; 9.78 at p 0.01; 23.7 at p 0.001

Latin Square Design

- **Result**
- There is very highly significant effect of fertilizer on the fresh weight of plants ($F_{3,6} = 311.79$, $p < 0.001$).
- The effect of road on the fresh weight of plants is non-significant ($F_{3,6} = 0.412$, $p > 0.05$).
- There is highly significant effect of river on the fresh weight of plants ($F_{3,6} = 14.928$, $p < 0.01$).